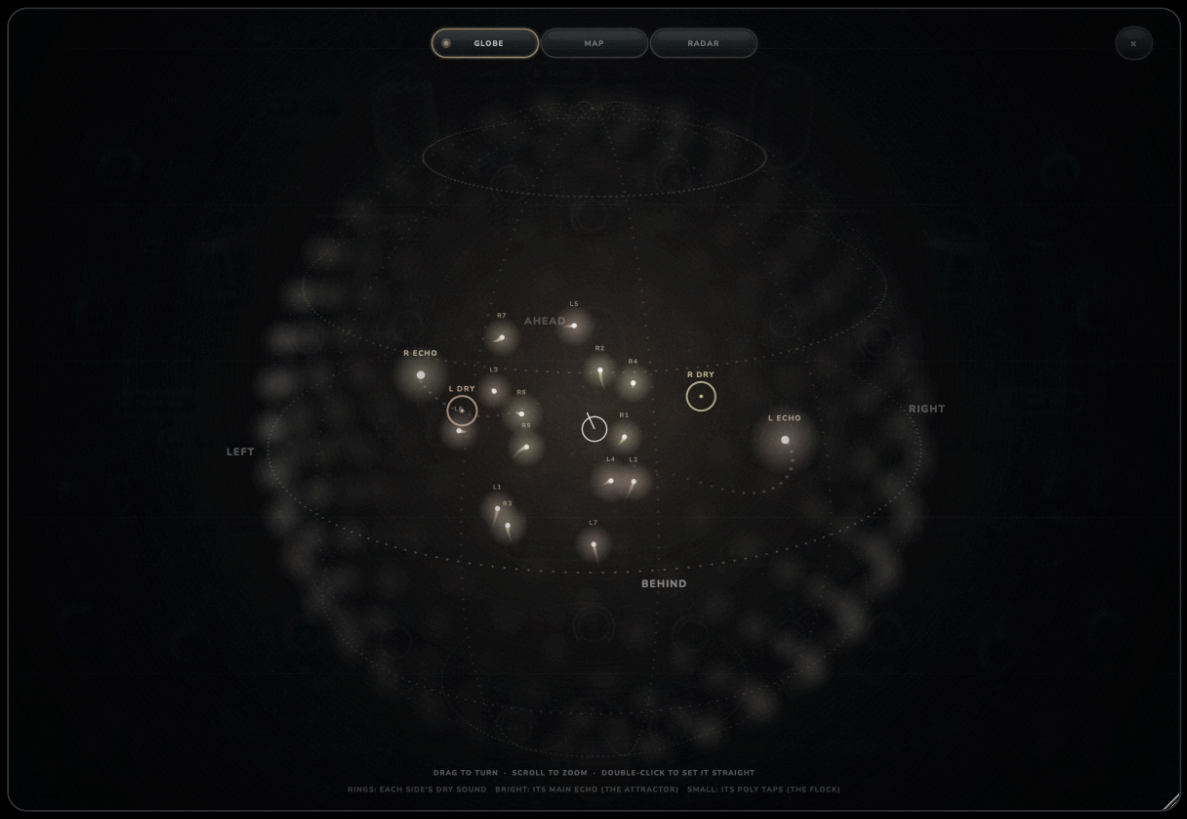
Chloaeilius

The spatial audio version of Chloaelis: an Ambisonics encoder (1st to 7th order), its echoes flying round you on the flock and the Lorenz attractor.

Manual for version 0.2.0



About this manual

This manual covers Chloaeilius 0.2.0, the plug-in (AU, VST3, LV2) and the desktop app. Chloaeilius is the same instrument as Chloaelis, so after the spatial chapters it carries Chloaelis's own: what it is, the layout, the symbols and how to play it. Then a reference for every parameter, and answers to common problems.

> **Spatial audio.** Chloaeilius is the Ambisonics version (1st to 7th order, 3rd by default) of Chloaelis, kept in the Chloaelis repository (this folder). It is an **encoder**: stereo goes in, Ambisonics come out (ACN order, SN3D, the format of the IEM and SPARTA plug-ins: 16 channels at 3rd order, 36 at 5th, 64 at 7th), and it decodes nothing itself. Route all of them to a decoder of the same order. For the ordinary stereo instrument, see Chloaelis.



The manual (PDF): the sphere, the desktop app, Chloaelis's controls, every parameter and answers to common problems.

The sphere

Chloaeilius is an encoder. It sends Ambisonics (ACN order, SN3D, the format of the IEM and SPARTA plug-ins) and decodes nothing itself. The **order** switch at the top left picks how sharp it is, from 1st to 7th order: 3rd (the default) is 16 channels, 5th is 36, 7th is 64. In REAPER, give the track that many channels and put the IEM AllRADecoder (loudspeakers) or BinauralDecoder (headphones) after it at the same order, SN3D. On a track with too few channels for the order you picked, the highest order that fits comes out. At 7th order all 18 points take about 14 % of one core, against 10 % at 3rd.

Where the sound goes:

- Each side's **dry sound** is a point: the left side ahead and to the left, the right side ahead and to the right.
- Each side's **main echo** rides the Lorenz attractor (the right side's mirrored through you), so the delay's repeats loop round you in the butterfly's figure.
- Each of the **seven poly taps** per side rides one bird of the flock, now flying in three dimensions round you. A feedback trail comes back from wherever its bird has flown, so repeats drift through the room, and the birds gather and scatter round you rather than to one side.
- **Distance**, after the IEM RoomEncoder: every point has its own propagation delay (so echoes flying towards you or away bend in pitch), 1/r level, a little air absorption and first-order reflections from the walls of a room round you. The **Distance** knob (lambda) is the size of the bubble you're in, from 1.5 m to 24 m: in a small bubble everything is close and loud and the room answers quickly; in a big one everything is far, quieter, later and roomier.
- With the order on *delay > filter*, the dry sound goes through the tube and filter as before and each tap through its own filter (the taps skip the tube there, so they stay apart).

Space View, the switch under the title, shows the sphere four ways (the sound is the same in all of them):

Lorenz: the attractor alone, as in Chloaelis



Orrery: a dotted globe round you, turning slowly and leaning with the cursor; every point flies inside it with its trail, and the loud places glow on its skin



Map: the whole sphere unrolled (a Hammer projection, as the IEM EnergyVisualizer), ahead in the middle, behind at both edges, with the energy as glowing dust



Radar: from above, rings in metres out to the bubble's edge, each point on a stalk as tall as it is high, and a ripple wherever an echo lands



The sphere button (the circle under **?** in the top right) opens a big sphere over the whole window, with nothing in the way: drag to turn it, scroll to zoom, double-click to set it straight, and switch between globe, map and radar along the top. Every point is named (L DRY, L ECHO, L1 to L7 for the taps, and the same for R), and the loud places glow on the glass.



The desktop app

`Chloaeilius.app` (`Chloaeilius.exe` on Windows, `Chloaeilius` on Linux) is Chloaeilius on its own, with no DAW, and it has Chloaelis's desktop extras:

- **The vessel** (top right): load your own sample. Drop an audio file onto it (WAV, AIFF, FLAC, MP3, M4A, OGG or CAF, up to 10 minutes) or double-click it to choose one. The two rings set the loop, and dragging the glass sets its level. It plays into both sides and its echoes fly round you like everything else.
- **Listen** (top left), because without a DAW there's nowhere to put a decoder:
  - **Headphones**: the sphere decoded to binaural (a spherical-head model), on outputs 1 and 2.
  - **Speakers**: a stereo decode for two loudspeakers.
  - **16 ch**: the Ambisonics themselves, for a multichannel interface or to route into another app (for example through BlackHole into REAPER and an IEM decoder).
- **Options** (the audio device and channels), **fullscreen** (cmd+F, esc to leave) and the help, as in Chloaelis.

The plug-ins (AU, VST3, LV2) have no Listen switch: in a DAW they always send the Ambisonics to your decoder. The desktop app works at 3rd order.



Build it

Chloaeilius isn't released as a download yet. When it is, its releases will be tagged `chloaeilius-v...` on the Chloaelis Releases page, separate from Chloaelis's own. For now, build it from this folder (it shares the repository's JUCE, fonts and recordings, so clone the whole thing):

```
git clone --recursive https://github.com/femifleming/Chloaelis.git
cd Chloaelis
cmake -S Chloaeilius -B build-chloaeilius -DCMAKE_BUILD_TYPE=Release -
DCHLOAEILIUS_COPY_AFTER_BUILD=ON
cmake --build build-chloaeilius --config Release --target Chloaeilius_All
```

It appears in your DAW as **Chloaeilius** under **Chloaelis**. The tests are `ChloaeiliusTest`, `ChloaeiliusSpatialTest` (where the sound lands, every channel against AmbiX, CPU) and `ChloaeiliusHostTest` (`SPACE_SHOTS=<folder>` saves every view of the sphere).

Everything else (the controls, the delay engines, the tube, the help) works as in Chloaelis: see its README.

Human contributions

Chloaeilius was written with Claude Code (Anthropic) under the direction of its author, femifleming. It is the spatial version of Chloaelis. The idea, the references (Bizarre Jezabel's *Chloe* and the game *Splice*), the look and the field recordings all come from the author's work on Chloaelis: see Human contributions in its README.

What it is (as in Chloaelis)

Chloaelis is two complete signal paths, hard-panned: the **left input goes through the left side to the left output**, and the right through the right. Each side runs

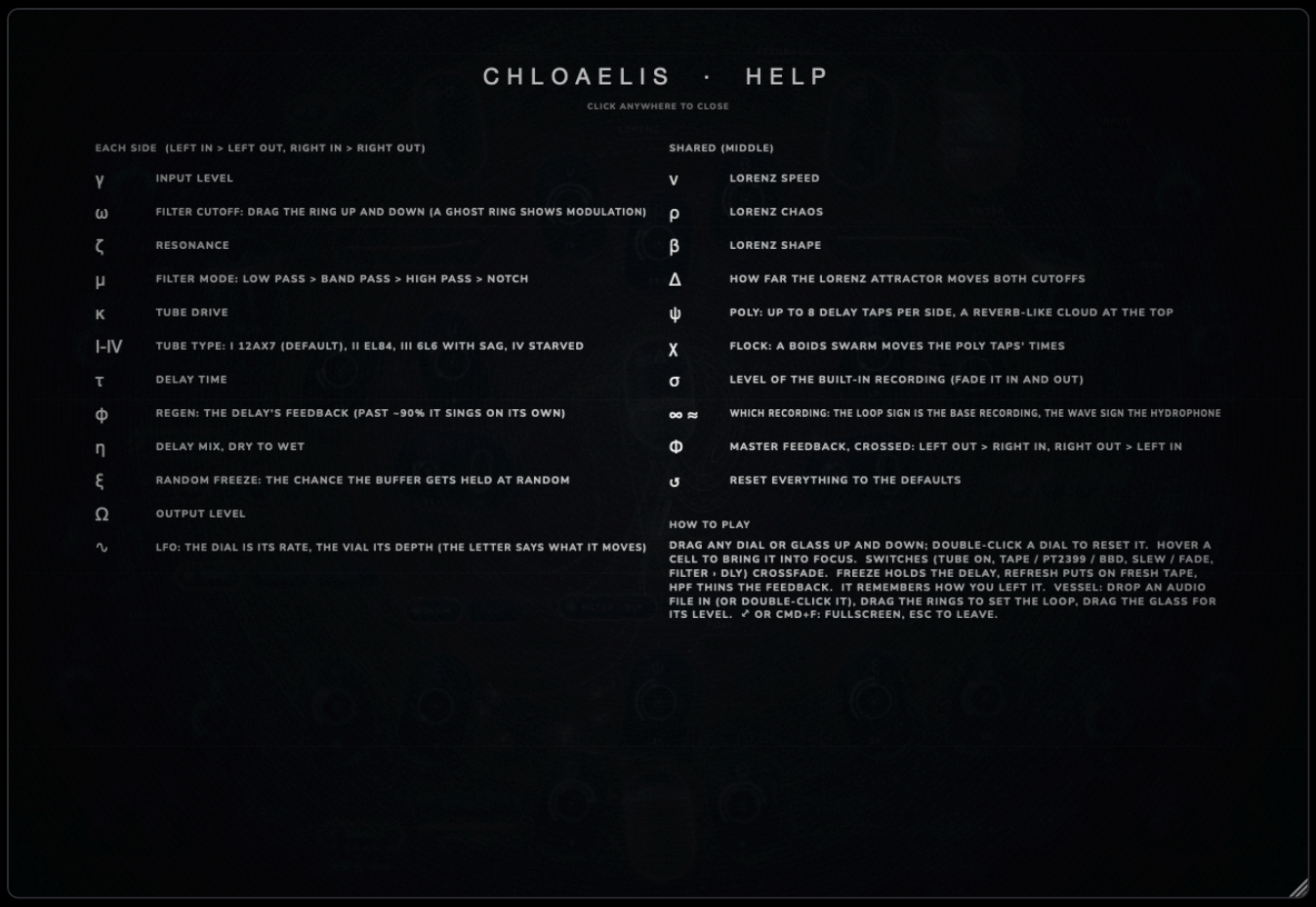
input > multimode filter > tube > delay > output

(or delay > tube > filter, with the order switch). Shared controls in the middle tie the two sides together:

- **Filter**: continuous low pass > band pass > high pass > notch, with resonance.
- **Tube**: four types, each oversampled 4x: I 12AX7 (the default: warm, two triode stages), II EL84 (push-pull pentodes, crunchy), III 6L6 (with power-supply sag, it blooms) and IV starved (a low-voltage 12AX7, gated and sputtery).
- **Delay engines**: *tape* (an ageing tape loop: hiss, wow and flutter, oxide wear; it ages while it runs and *refresh* puts on new tape), *PT2399* (the delay chip, gritty at long times) and *BBD* (bucket brigade, dark and crunchy). Up to 1 s, regen that self-oscillates, a feedback high-pass, freeze, random freeze, and slew (pitch-bending) or fade time changes.
- **Poly**: fades in up to 8 delay taps per side at irregular offsets. All the way up it becomes a reverb-like cloud. **Flock** lets a boids simulation move those taps' times, so they drift, cluster and scatter.
- **Lorenz attractor**: its x and y move the left and right cutoffs.
- **LFOs**: triangle LFOs on filter mode, delay time and tube drive, per side.
- **Master feedback**, crossed over: the left output feeds the right input and the right output the left.
- **Two built-in recordings** to play through it: a long loop from *baserecording1* and a hydrophone recording. Fade them in and out; the left channel goes to the left side and the right to the right.
- Every switch (order, delay engine, time mode, tube type) **crossfades**: echoes ring out in the old setting while new sound goes through the new one, so there are no cuts.
- It **remembers** how you left it, and ↺ resets everything.

How to use it (as in Chloaelis)

The layout



- The **left half** of the window is the left channel and the **right half** is the right channel. They mirror each other: input at the top, then filter, tube and LFOs, delay at the bottom, output at the bottom middle.
- The **middle** holds the shared controls: the Lorenz attractor (top), the poly and flock (bottom), the order switch, and the built-in recordings, master feedback and the vessel (top row).
- **Glowing dashed lines** show where signal and modulation flow. Pulses of light follow the signal.
- Cells **glow with level**: the side's colour when it's moving, warm white when it's hot and red when it's close to clipping.
- The **Lorenz attractor** floats in the background as a hologram, with dashed lines to the two cutoffs. The **boids** of the flock fly around the poly controls.

The symbols

Every control is named by a Greek letter. Press **?** (top right) for this list inside the app.

Each side

Symbol	Control	What it does
γ	Input	input level
ω	Cutoff	filter cutoff: drag the ring up and down. A faint ring shows where modulation has pushed it
ζ	Resonance	filter resonance

μ	Mode	filter mode: low pass > band pass > high pass > notch, continuously
κ	Drive	tube drive
I–IV	Tube type	I 12AX7 (default), II EL84, III 6L6 with sag, IV starved
TUBE ON		switches the tube in and out
τ	Time	delay time, 30 ms to 1 s
φ	Regen	the delay's feedback. Past about 90% it sings on its own
η	Mix	delay dry / wet
ξ	Random freeze	the chance that the delay buffer gets held for a while at random moments
TAPE / PT2399 / BBD		the delay engine
SLEW / FADE		time changes either glide (pitch bends) or crossfade (no pitch change)
HPF		high-pass filter in the delay feedback
FREEZE		holds the delay buffer. On tape, a held loop slowly wears away
REFRESH		puts on fresh tape (tape engine only)
Ω	Out	output level
∿	LFO	the dial is its rate, the small glass vial its depth. The letter next to ∿ says what it moves (μ mode, τ time, κ drive)

Shared (middle)

Symbol	Control	What it does
ν	Speed	how fast the Lorenz attractor moves
ρ	Chaos	the attractor's rho: low wanders gently, high is wild
β	Shape	the attractor's beta: reshapes the orbit
Δ	Depth	how far the attractor moves the two cutoffs (up to ±3.5 octaves)
FILTER › DLY / DLY › FILTER	Order	filter > tube > delay, or delay > tube > filter
ψ	Poly	up to 8 delay taps per side. All the way up is a reverb-like cloud
χ	Flock	how much the boids move the poly taps' times (up to about ±0.9 octave)
∞ / ≈	Source	which built-in recording: ∞ the base loop, ≈ the hydrophone
σ	Source level	fades the built-in recording in and out
Φ	Master feedback	crossed: left out > right in, right out > left in
↺	Reset	every control back to its default (the tape's wear is kept)

?	Help	the key to all of this, inside the app

Playing it

- **Drag up and down** on any dial, glass vial or ring to change it. **Double-click** a dial to reset it to its default.
- **Hover** over a control to bring it into focus. The depth of field follows you, and the field tilts slightly toward the cursor.
- **No input?** Turn up **σ** to play one of the built-in recordings through it, and pick which with **∞ / ≈**.
- **Ideas to start with**
  - *Ambient wash*: σ up, ψ (poly) and χ (flock) up, φ (regen) around 60%, tape engine.
  - *Lo-fi echo*: PT2399 or BBD, τ around 400 ms, φ around 70%, κ (drive) up a little.
  - *Moving filter*: Δ (Lorenz depth) up, ρ (chaos) up, ζ (resonance) up.
  - *Self-playing*: Φ (master feedback) towards 90% with some regen: the two sides feed each other and keep going on their own.
- Chloaelis **remembers its last state**: a new instance starts how the last one was closed. In a DAW, the project's own saved settings take priority.

The desktop app

The app (`Chloaelis.app`, `Chloaelis.exe` or `Chloaelis` on Linux) has a few extras:

- **Audio settings**: the **Options** button at the top left of the window chooses the audio device, the input and output channels and the buffer size.
- **Fullscreen**: the ⤢ button, or **⌘F** (Ctrl+F on Windows and Linux). It fills the whole screen with no menu bar or title bar, at any screen shape. **Esc** leaves it.
- **The vessel** (top right): a glass you can pour your own audio into.
  - **Drop an audio file** onto it (WAV, AIFF, FLAC, MP3, M4A, OGG or CAF, up to 10 minutes), or **double-click** it to choose one.
  - Its waveform runs up the inside, start at the bottom.
  - The **two rings** are the loop start and end: drag them.
  - **Drag the glass** itself (away from the rings) to set its level, shown by the liquid.
  - It plays into both sides like the built-in recordings, and is remembered with the rest of the state.

Help inside the app

Press **?** (top right) for a full key to the symbols and a short how-to. Click anywhere, or press Esc in the app, to close it.

Parameter reference

Every control your DAW can automate, by the name it shows there. The two sides have the same controls, so the right side's table follows the left's. Defaults are what a fresh instance starts with (reset brings them back).

Shared (the middle)

Name in your DAW	ID	Range	Default
Lorenz Speed	`lz_speed`	0% to 100%	35%
Lorenz Chaos	`lz_chaos`	ρ 24.5 to ρ 180.0	ρ 38.5
Lorenz Shape	`lz_shape`	β 0.50 to β 4.00	β 2.67
Lorenz > Cutoff	`lz_depth`	0% to 100%	0%
Poly Width	`width`	OFF to 100%	OFF
Poly Flock	`poly_flock`	0% to 100%	0%
Order	`order`	Filter > Delay, Delay > Filter	Filter > Delay
Source	`src_sel`	Base Loop, Hydrophone	Base Loop
Source Level	`src_level`	0% to 100%	0%
Master Feedback	`master_fb`	0% to 100%	0%
Distance	`space`	1.5 m to 24.0 m	7.9 m
Listen (desktop app)	`listen`	Headphones, Speakers, 16 ch	Headphones
Space View	`space_view`	Lorenz, Orrery, Map, Radar	Orrery
Ambisonic Order	`ambi_order`	1st, 2nd, 3rd, 4th, 5th, 6th, 7th	3rd

Left side

Name in your DAW	ID	Range	Default
L In Att	`l_inatt`	silent to 12.0 dB	0.0 dB
L Cutoff	`l_cutoff`	20 Hz to 20.0 kHz	20.0 kHz
L Resonance	`l_res`	0% to 100%	15%
L Mode	`l_morph`	LP to Notch	LP
L Drive	`l_drive`	0% to 100%	30%
L Time	`l_time`	30 ms to 1000 ms	330 ms
L Regen	`l_regen`	0% to 115%	40%
L Wet	`l_wet`	0% to 100%	35%
L Out	`l_out`	silent to 12.0 dB	0.0 dB
L Random Freeze	`l_randfrz`	0% to 100%	0%
L LFO>Mode Rate	`l_lfomode_rate`	0.02 Hz to 20.0 Hz	0.21 Hz
L LFO>Mode Depth	`l_lfomode_depth`	0% to 100%	0%
L LFO>Time Rate	`l_lfotime_rate`	0.02 Hz to 20.0 Hz	0.11 Hz
L LFO>Time Depth	`l_lfotime_depth`	0% to 100%	0%
L LFO>Drive Rate	`l_lfodrive_rate`	0.02 Hz to 20.0 Hz	0.37 Hz

L LFO>Drive Depth	`l_lfodrive_depth`	0% to 100%	0%
L Delay HPF	`l_hpf`	off / on	Off
L Freeze	`l_freeze`	off / on	Off
L Tube On	`l_tubeon`	off / on	On
L Delay Type	`l_dtype`	Tape, PT2399, BBD	PT2399
L Time Change	`l_tmode`	Slew, Fade	Slew
L Tube Type	`l_tubetype`	12AX7, EL84, 6L6, Starved	12AX7

Right side

Name in your DAW	ID	Range	Default
R In Att	`r_inatt`	silent to 12.0 dB	0.0 dB
R Cutoff	`r_cutoff`	20 Hz to 20.0 kHz	20.0 kHz
R Resonance	`r_res`	0% to 100%	15%
R Mode	`r_morph`	LP to Notch	LP
R Drive	`r_drive`	0% to 100%	30%
R Time	`r_time`	30 ms to 1000 ms	410 ms
R Regen	`r_regen`	0% to 115%	40%
R Wet	`r_wet`	0% to 100%	35%
R Out	`r_out`	silent to 12.0 dB	0.0 dB
R Random Freeze	`r_randfrz`	0% to 100%	0%
R LFO>Mode Rate	`r_lfomode_rate`	0.02 Hz to 20.0 Hz	0.17 Hz
R LFO>Mode Depth	`r_lfomode_depth`	0% to 100%	0%
R LFO>Time Rate	`r_lfotime_rate`	0.02 Hz to 20.0 Hz	0.13 Hz
R LFO>Time Depth	`r_lfotime_depth`	0% to 100%	0%
R LFO>Drive Rate	`r_lfodrive_rate`	0.02 Hz to 20.0 Hz	0.29 Hz
R LFO>Drive Depth	`r_lfodrive_depth`	0% to 100%	0%
R Delay HPF	`r_hpf`	off / on	Off
R Freeze	`r_freeze`	off / on	Off
R Tube On	`r_tubeon`	off / on	On
R Delay Type	`r_dtype`	Tape, PT2399, BBD	PT2399
R Time Change	`r_tmode`	Slew, Fade	Slew
R Tube Type	`r_tubetype`	12AX7, EL84, 6L6, Starved	12AX7

Distance is the radius of the bubble you're in. Ambisonic Order sets how many channels go out (16 at 3rd order, 36 at 5th, 64 at 7th). Listen only does anything in the desktop app; the plug-ins always send the Ambisonics. Space View changes the picture, never the sound.

Problems and answers

I only hear two channels, or nothing moves round me

In a DAW the track needs as many channels as the order switch says (16 at 3rd order, 36 at 5th, 64 at 7th) and a decoder after Chloaeilius at the same order (ACN, SN3D), such as the IEM AllRADecoder or BinauralDecoder. In the desktop app, set Listen to Headphones or Speakers.

macOS says it can't be opened, or the DAW doesn't list it

The builds aren't notarised. Run `xattr -dr com.apple.quarantine` on the app and the plug-ins (see Installing), then rescan in your DAW.

There's no sound and nothing on the input

Turn up σ (source level) to play a built-in recording through it, and pick which with ∞ / ≈. In the desktop app, check Options for the audio device.

It keeps sounding after I stop

Regen past about 90 % sings on its own, and master feedback (Φ) keeps the two sides feeding each other. Turn them down, or reset with ↺.

A new instance doesn't start from the defaults

It remembers its last state, so a new instance starts how the last one was closed. A saved project still opens as you saved it. ↺ resets everything (the tape's wear is kept).